

Project Documentation

1. Introduction

Project Title: Script-Controlled ACL - Restrict Record Access Based on Field Value

Team Members:

Name	Role
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2. Project Overview

Purpose: To restrict who can view, create, edit and delete records of a custom table by combining user roles, a data condition on the Branch field and an advanced ACL script. Only users with the right role see the right records, and administrators keep full access.

Features:

- Custom table Institution Details (u_institution_details) with student, faculty, branch, email, phone and description fields.
- Custom roles bb1 (read), bb2 (create), bb3 (write) and bb4 (delete).
- READ ACL with data condition Branch is EEE and an advanced script.
- Separate CREATE, WRITE and DELETE ACLs.
- Admin override so administrators see all records.

3. Architecture

- **Frontend (React equivalent):** The ServiceNow platform UI. Users work in the standard list and form views of the Institution Details table. The UI adapts to the user's ACLs: rows are filtered, the New button appears only with bb2, fields are editable only with bb3 and Delete appears only with bb4.
- **Backend (Node.js and Express.js equivalent):** ServiceNow server-side logic. Each request is checked by the matching ACL, using the role requirement, the data condition and the advanced script (server-side JavaScript with gs.hasRole()).
- **Database (MongoDB equivalent):** The ServiceNow database table u_institution_details.

Field	Type	Notes
Student Roll Number	Auto number	Default INS000xxxx
Student Name	Reference (User)	
Faculty Name	Reference (User)	
Branch	Choice	ECE, EEE, CSE
Email	String	
Phone Number	String	
Description	Multi-line string	Editable with bb3

4.Setup Instructions

Prerequisites:

- A ServiceNow instance (for example a Personal Developer Instance) and a login with admin access.
- Ability to elevate to the security_admin role, needed to create ACLs.
- A modern web browser. No local software, Node.js or MongoDB installation is needed.

Installation (no cloning or environment variables are needed):

- Sign in to the instance as admin.
- Create the user EEE User (User Administration > Users).
- Create roles bb1, bb2, bb3, bb4 and assign them to EEE User.
- Create the table u_institution_details and add its fields, then enter sample records for ECE, EEE and CSE.
- Elevate the security_admin role, then create the READ, CREATE, WRITE and DELETE ACLs (System Security > Access Control (ACL)).
- Capture the work in an update set (the lab used the update set "ACL Project [Global]") so it can be moved to another instance.

5.Folder Structure

Method	Endpoint	Purpose
GET	/api/now/table/u_institution_details	List records the caller may read

Client: Not applicable. There are no React source files; the UI is provided by the platform.

Server: Not applicable. There is no Node.js code; the configuration is stored as ServiceNow records:

Item	Where it lives
Users and roles	User Administration > Users, Roles
Table and fields	System Definition > Tables
ACLs and script	System Security > Access Control (ACL)
Packaging	Update set "ACL Project [Global]", Global scope

6. Running the Application

There are no npm start commands. To run and test the project:

- **Frontend:** sign in to the instance and open the list by entering `u_institution_details.list` in the All menu filter.
- **Backend:** nothing to start; the ACLs run automatically on every request.
- Test as different users with Impersonate user (EEE User, a user without the roles, and Admin).

7. API Documentation

This project adds no custom endpoints. ServiceNow's built-in REST Table API exposes the table, and the ACLs apply to it as well. The examples below are illustrative; check your instance for exact field names.

Method	Endpoint	Purpose
GET	<code>/api/now/table/u_institution_details/{sys_id}</code>	Get one record
POST	<code>/api/now/table/u_institution_details</code>	Create a record (needs bb2)
PATCH	<code>/api/now/table/u_institution_details/{sys_id}</code>	Update a record (needs bb3)
DELETE	<code>/api/now/table/u_institution_details/{sys_id}</code>	Delete a record (needs bb4)

Parameters: `sysparm_query` (for example `u_branch=EEE`), `sysparm_limit`, `sysparm_fields`.

Example response for GET (illustrative), as a user with bb1:

```
{ "result": [
  { "sys_id": "",
    "u_student_roll_number": "INS0001007",
    "u_branch": "EEE",
    "u_email": "saiguna@gmail.com",
    "u_phone_number": "9090898765" } ] }
```

A user without the required role receives an empty result list or an access-denied error, depending on the ACL that fails.

8.Authentication

- **Authentication:** Users sign in to the ServiceNow instance, which creates a session. API clients authenticate with their own credentials (for example basic auth or OAuth).
- **Authorization:** Roles bb1 to bb4 control each operation. For every request ServiceNow evaluates the matching ACL: the role requirement, the data condition (Branch is EEE) and the advanced script.
- **Admin handling:** Admins see all records through the script check and the Admin overrides setting.
- **Elevated privileges:** Creating ACLs requires elevating the security_admin role for the session.
- **Testing identities:** The Impersonate user feature switches between EEE User, a user without the roles, and Admin.

9.User Interface

institution Details

Student Roll Numbe

Search

Actions on selected rows...

All

Student Roll Number

Branch

Description

Email

Faculty name

Phone Number

Student Name

INS0001007

EEE

saiguna@gmail.com

Abel Tuter

9090898765

Guna Sai

INS0001004

EEE

prashanth@gmail.com

Abel Tuter

9090909090

Prashanth Reddy

User with bb1 role: only EEE records are listed

institution Details

Student Roll Numbe

Search

Student Roll Number

Branch

Description

Email

Faculty name

Phone Number

Student Name

Search

Search

Search

Search

Search

Search

Search



User without the role: no records are displayed

institution Details

Student Roll Numbe

Search

Actions on selected rows...

New

All

☐	Student Roll Number	Faculty name	Student Name	Branch	Email	Phone Number
	Search	Search	Search	Search	Search	Search
	INS0001006	ITIL User	Praneeth	ECE	praneeth@gmail.com	8989090906
	INS0001008	Malla Sravan kumar	Abraham Lincoln	CSE	sravan@gmail.com	8978908989
	INS0001005	ITIL User	Srinu Nandipalli	ECE	srinu@gmail.com	9089909090
	INS0001007	Abel Tuter	Guna Sai	EEE	saiguna@gmail.com	9090898765
	INS0001001	Abraham Lincoln	Yaswanth Nikku	CSE	yaswanth@gmail.com	8989898989
	INS0001004	Abel Tuter	Prashanth Reddy	EEE	prashanth@gmail.com	9090909090

Admin: all records are displayed

Student Roll Number	Branch	Description	Email	Faculty name	Phone Number	Student Name
INS0001007	EEE		salguna@gmail.com	Abel Tuter	9090898765	Guna Sai
INS0001004	EEE		prashanth@gmail.com	Abel Tuter	9090909090	Prashanth Reddy

User with bb1 and bb2: EEE records with the New button

10. Testing

Strategy: Manual role-based testing using impersonation, covering positive and negative cases for each ACL.

Tools: ServiceNow user impersonation, list views and the ACL forms.

Test	Result
bb1 user sees only EEE records (2 of 6)	Pass
User without the role sees no records	Pass
Admin sees all 6 records	Pass
bb1 + bb2: New button shown	Pass
Test	Result
bb1 to bb3: Description edited	Pass
bb1 to bb4: Delete option and confirmation shown	Pass
Negative checks for bb2, bb3, bb4 (without the role)	Pending
Performance: load time, concurrency, bulk load	Pending

11. Screenshots or Demo

Data Condition

Condition: No. of records matching the condition: 2

Add Filter Condition Add OR Clause

Branch is EEE

AND OR X

Advanced Condition

Controlled by References

Script

```

1
2
3 {function() {
4   if (gs.hasRole('bb1')) {
5     return true;
6   } else {
7     return false;
8   }
9 }}();

```

READ ACL: data condition (Branch is EEE) and script

Table Columns for text Search 1 to 13 of 13

Dictionary Entries

Column label	Type	Reference	Max length	Default value
Updates	Integer	(empty)	40	
Created by	String	(empty)	40	
Student Roll Number	String	(empty)	40	javascript:global.getNextObjNumberPadded();
Description	String	(empty)	1,000	
Updated by	String	(empty)	40	
Phone Number	String	(empty)	40	
Sys ID	Sys ID (GUID)	(empty)	32	
Created	Date/Time	(empty)	40	
Faculty name	Reference	User	32	
Branch	String	(empty)	40	
Updated	Date/Time	(empty)	40	
Student Name	Reference	User	32	
Email	String	(empty)	40	
Insert a new row...				

Table columns of u_institution_details

institution Details Student Roll Number Search Actions on selected rows... New

All

Student Roll Number	Branch	Description	Email	Faculty name	Phone Number	Student Name
INS0001007	EEE	ACL is Executed	saiguna@gmail.com	Abel Tuter	9090898765	Guna Sai
INS0001004	EEE		prashanth@gmail.com	Abel Tuter	9090909090	Prashanth Reddy

bb3 user editing the Description field

institution Details Student Roll Number Search Actions on selected rows...

All

Student Roll Number	Branch	Description	Faculty name	Phone Number	Student Name
INS0001007	EEE	ACL is Executed	Tuter	9090898765	Guna Sai
INS0001004	EEE	ACL is Executed	Tuter	9090909090	Prashanth Reddy

Confirmation
Delete the selected record(s)?
Cancel Delete

bb4 user deleting a record (confirmation)

Demo link:

12.Known Issues

ID	Issue	Severity
BG-001	Branch appears as a String field in the columns screenshot, but the instructions say Choice (ECE, EEE, CSE).	Low
BG-002	The READ ACL script in the screenshot checks only bb1, while the instructions also include an admin check. Admins still see all records because of Admin overrides.	Low
BG-003	The ACL logic is tied to EEE and bb1; other branches are not yet covered.	Low

13.Future Enhancements

- Add roles and conditions for ECE and CSE.
- Add field-level ACLs for Email and Phone Number.
- Use groups or a branch lookup instead of hard-coded roles.
- Extend the ACLs to Service Portal pages.
- Add automated tests, for example with ServiceNow's Automated Test Framework.